

UNIVERSIDAD NACIONAL DE CÓRDOBA
FACULTAD DE CIENCIAS EXACTAS, FÍSICAS Y NATURALES

TEORÍA DE SEÑALES Y SISTEMAS LINEALES

PRÁCTICO: Problemas Capitulo 2 CONVOLUCION DE SEÑALES EN TIEMPO DISCRETO

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SEÑALES y SISTEMAS DE TIEMPO DISCRETO

BIBLIOGRAFÍA

*** Señales y Sistemas, Segunda Edición A.
V. Oppenheim, A. S. Willsky, S. H. Nawab.
Prentice Hall**

2.21. Calcule la convolución $y[n] = x[n] * h[n]$ de los siguientes pares de señales:

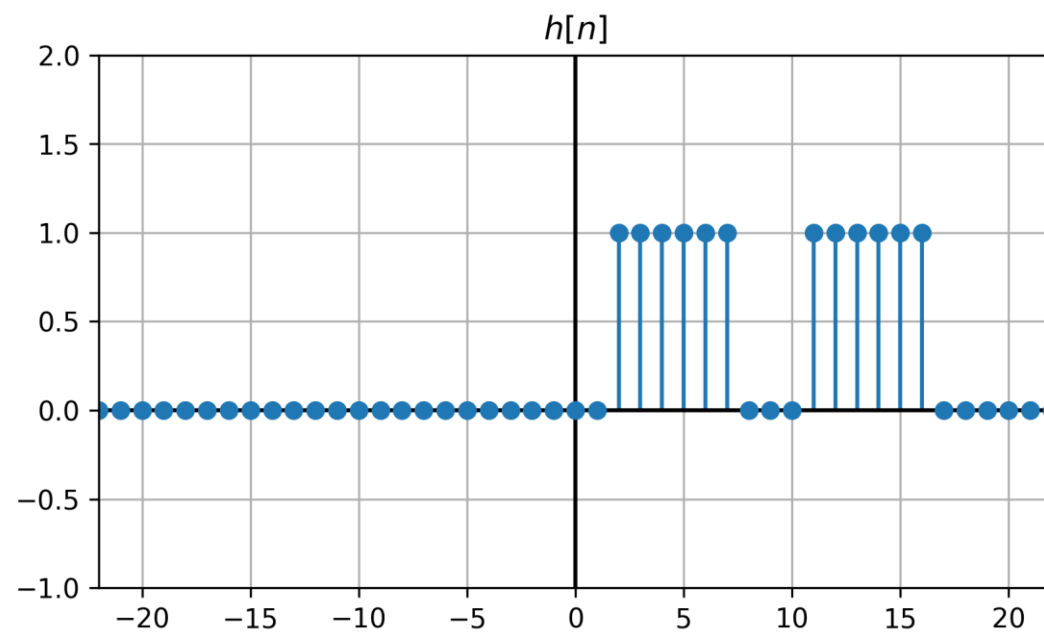
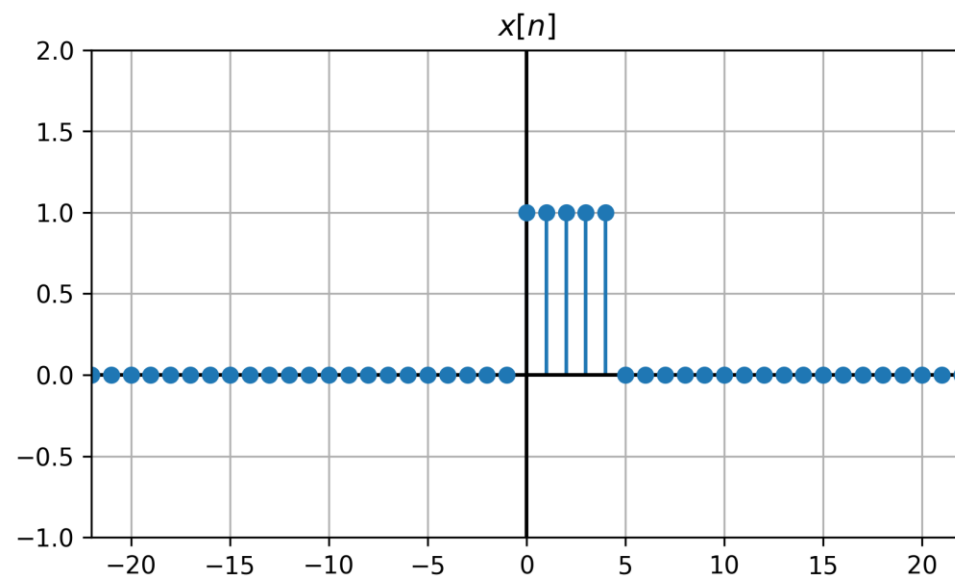
$$a) \ x[n] = \alpha^n u[n]$$

$$\alpha \neq \beta$$

$$h[n] = \beta^n u[n]$$

$$b) \ x[n] = h[n] = \alpha^n u[n]$$

d) $x[n]$ y $h[n]$ son como en la figura:



$$a) \ x[n] = \alpha^n u[n]$$

$$\alpha \neq \beta$$

$$h[n] = \beta^n u[n]$$

Solución:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} \alpha^k u[k] \beta^{n-k} u[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} \alpha^k u[k] \frac{\beta^n}{\beta^k} u[n-k]$$

$$y[n] = \beta^n \sum_{k=-\infty}^{\infty} \left(\frac{\alpha}{\beta}\right)^k u[k] u[n-k]$$

$$y[n] = \beta^n \sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k \quad ; \quad n \geq 0$$

Calculo auxiliar:

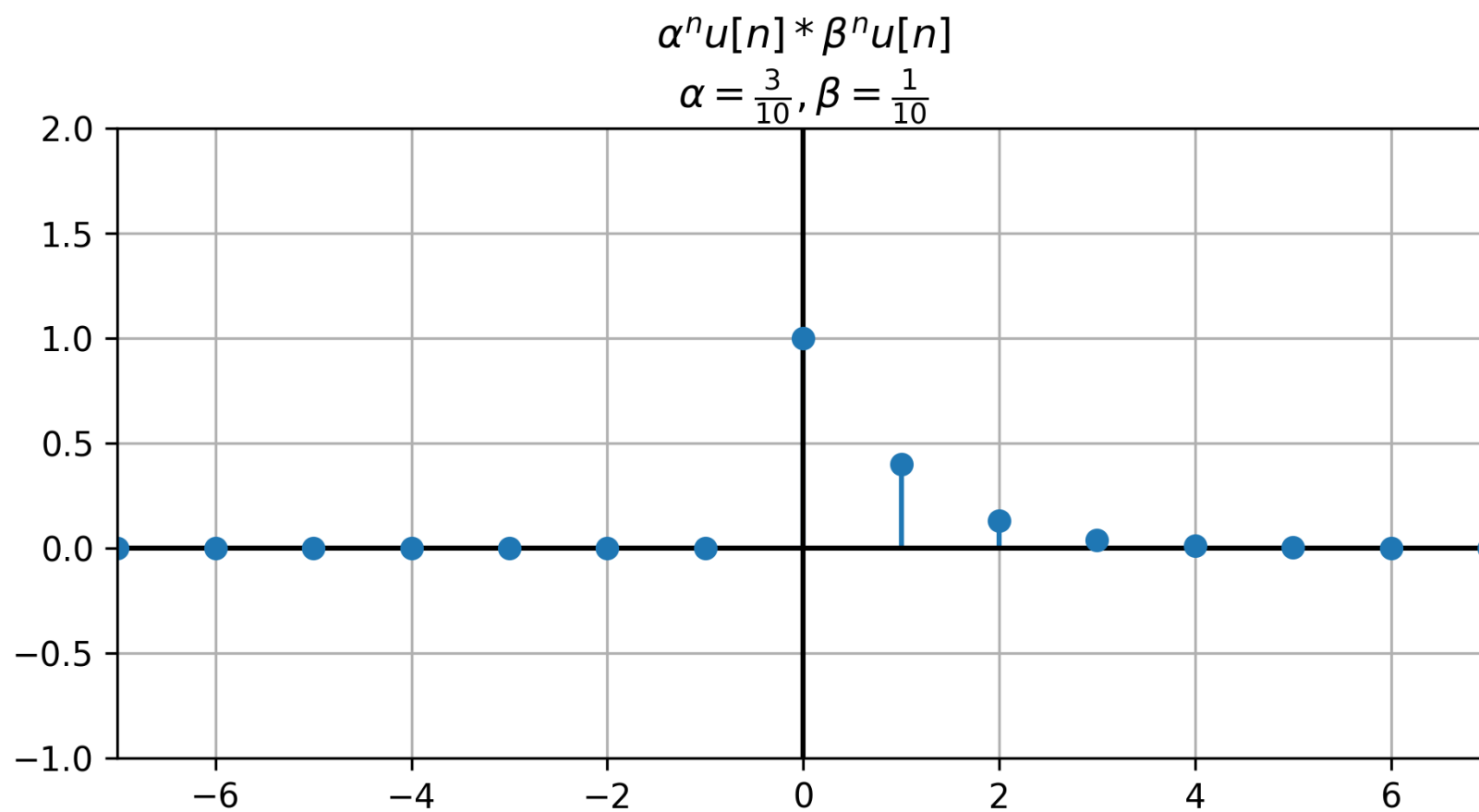
Serie Geométrica

$$\sum_{k=0}^{m-1} r^k = \frac{1 - r^m}{1 - r} \quad ; \quad r \neq 1$$

Ejemplo:

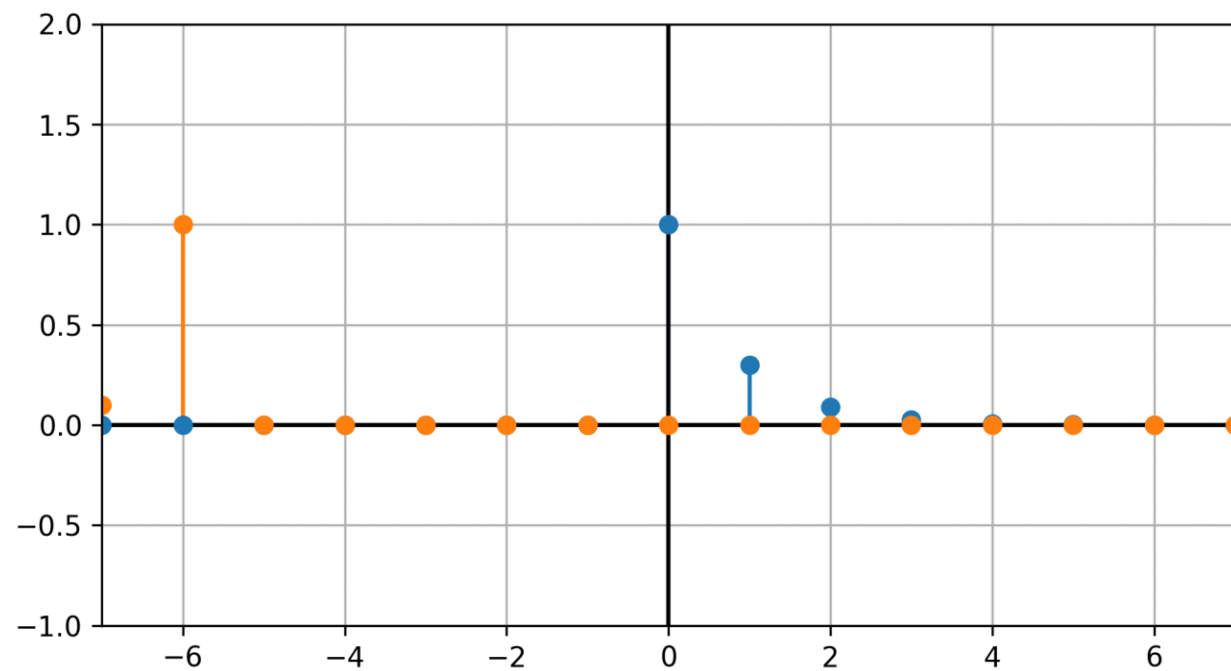
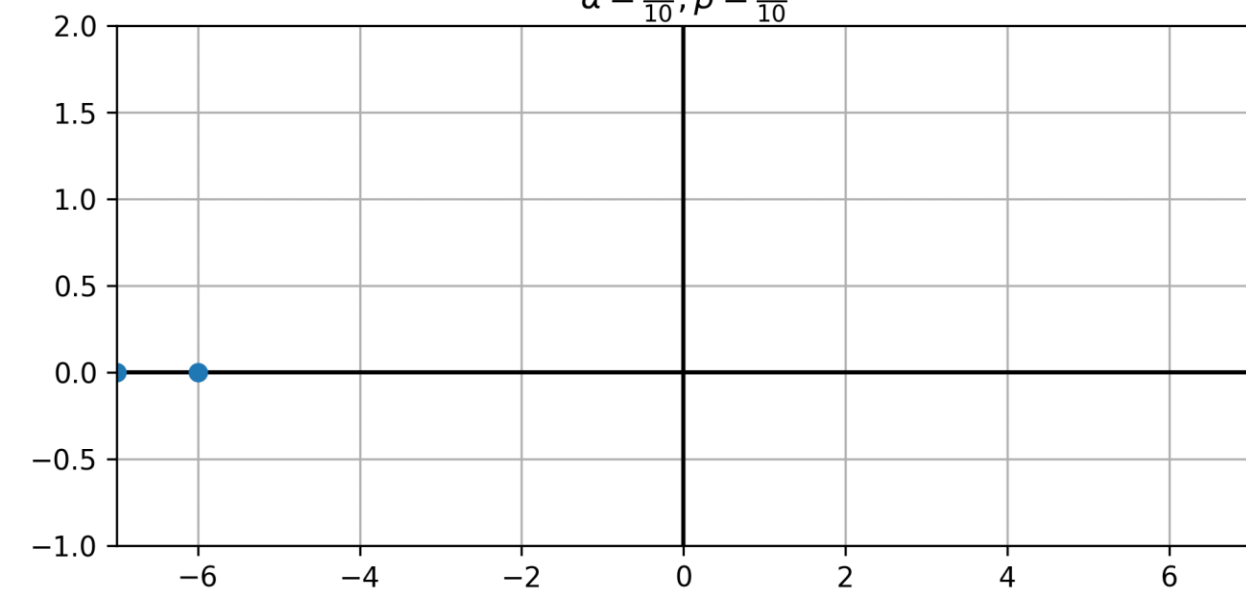
$$\begin{aligned}\sum_{k=0}^n \left(\frac{\alpha}{\beta}\right)^k &= \frac{1 - \left(\frac{\alpha}{\beta}\right)^{n+1}}{1 - \frac{\alpha}{\beta}} = \frac{1 - \frac{\alpha^{n+1}}{\beta^{n+1}}}{1 - \frac{\alpha}{\beta}} = \frac{\frac{\beta^{n+1} - \alpha^{n+1}}{\beta^{n+1}}}{\frac{\beta - \alpha}{\beta}} \\&= \frac{(\beta^{n+1} - \alpha^{n+1}) \beta}{\beta^{n+1} (\beta - \alpha)} = \frac{(\beta^{n+1} - \alpha^{n+1}) \beta}{\beta^n \beta (\beta - \alpha)} \\&= \frac{(\beta^{n+1} - \alpha^{n+1})}{\beta^n (\beta - \alpha)}\end{aligned}$$

$$y[n] = \left(\frac{\beta^{n+1} - \alpha^{n+1}}{\beta - \alpha} \right) u[n] \quad ; \quad \alpha \neq \beta$$



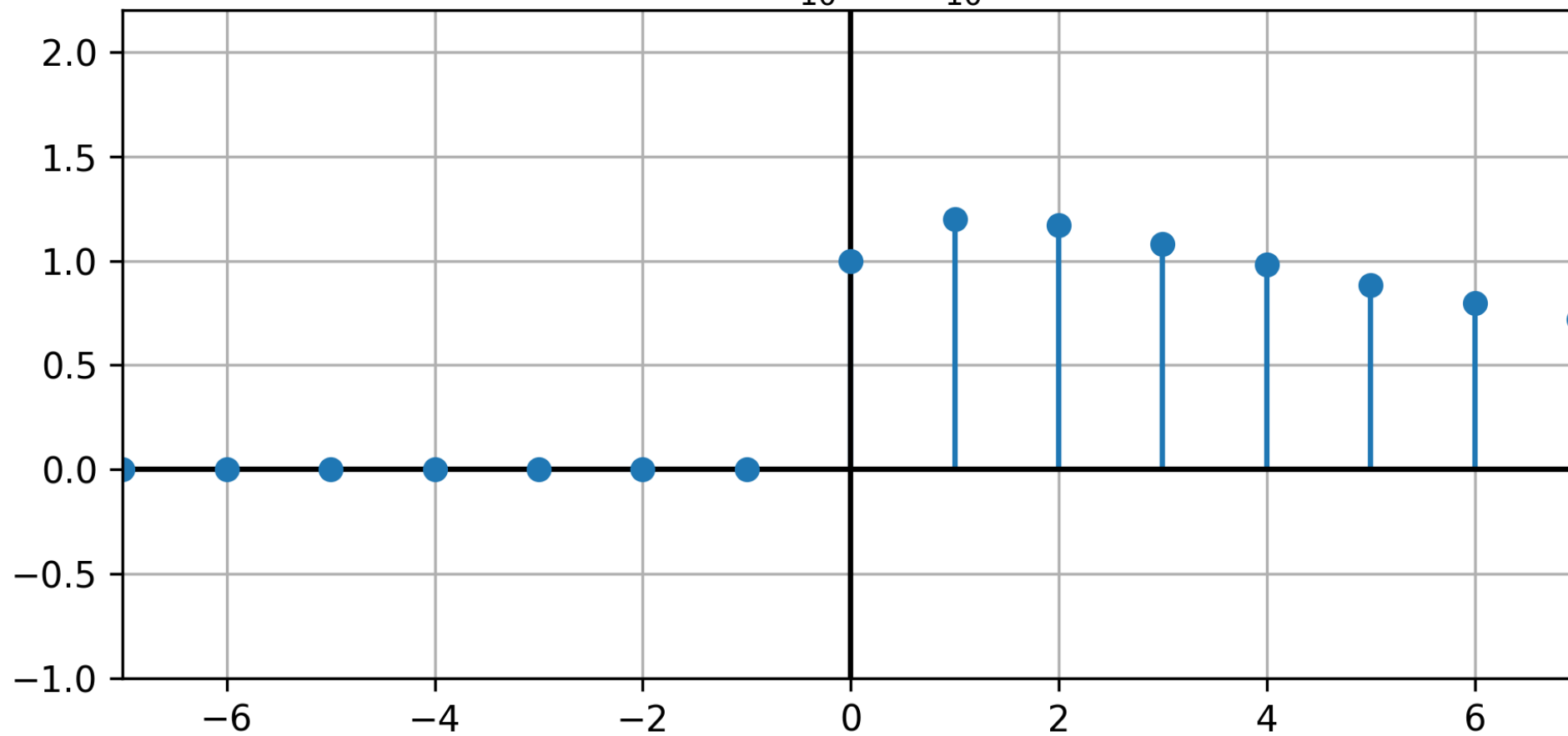
$$\alpha^n u[n] * \beta^n u[n]$$

$$\alpha = \frac{3}{10}, \beta = \frac{1}{10}$$



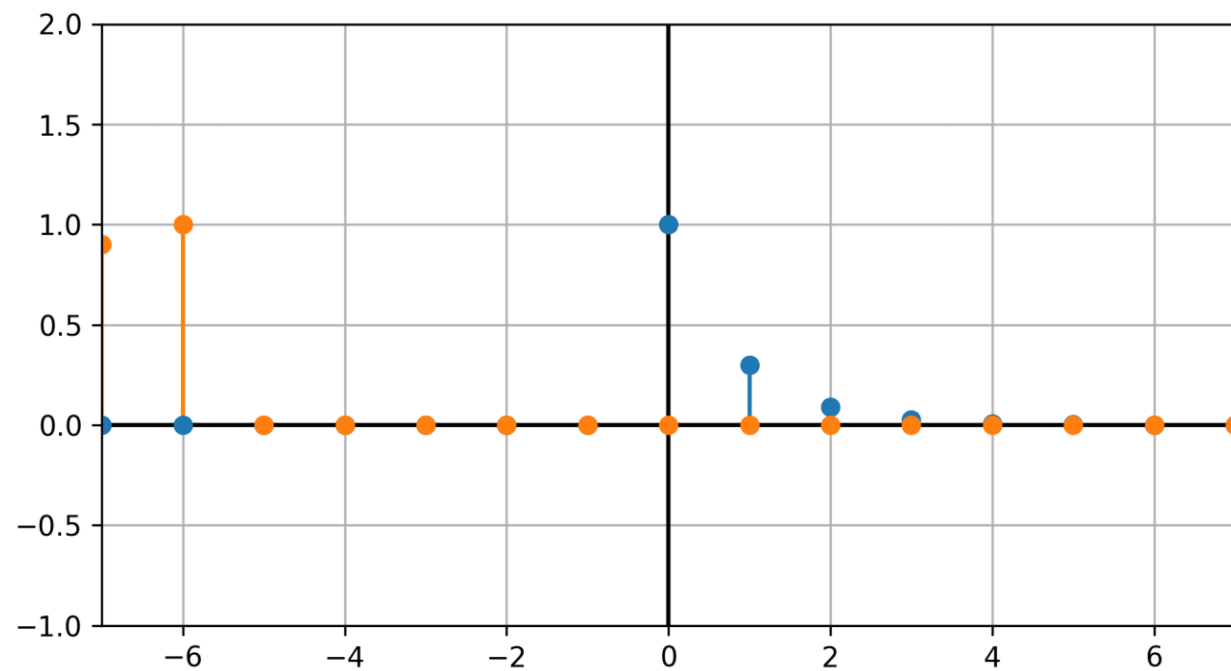
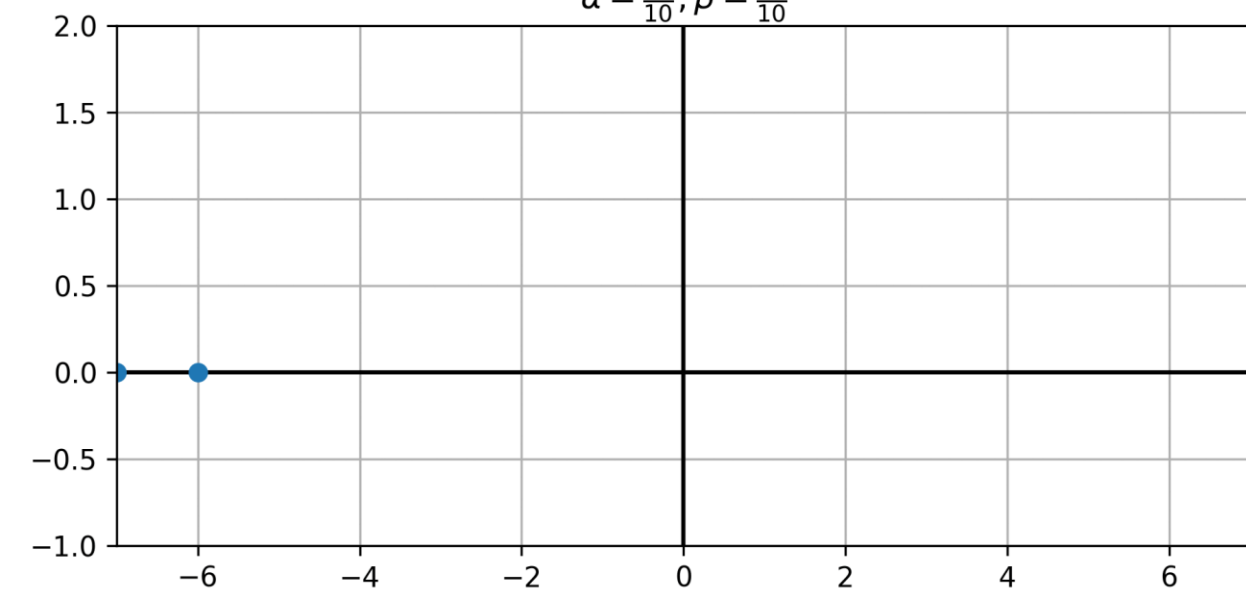
$$\alpha^n u[n] * \beta^n u[n]$$

$$\alpha = \frac{3}{10}, \beta = \frac{9}{10}$$



$$\alpha^n u[n] * \beta^n u[n]$$

$$\alpha = \frac{3}{10}, \beta = \frac{9}{10}$$



$$\text{b) } x[n] = h[n] = \alpha^n u[n]$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Calculo auxiliar:

$$x[k] = \alpha^k u[k]$$

$$h[n-k] = \alpha^{n-k} u[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} \alpha^k u[k] \alpha^{n-k} u[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} \alpha^k u[k] \alpha^{n-k} u[n-k]$$

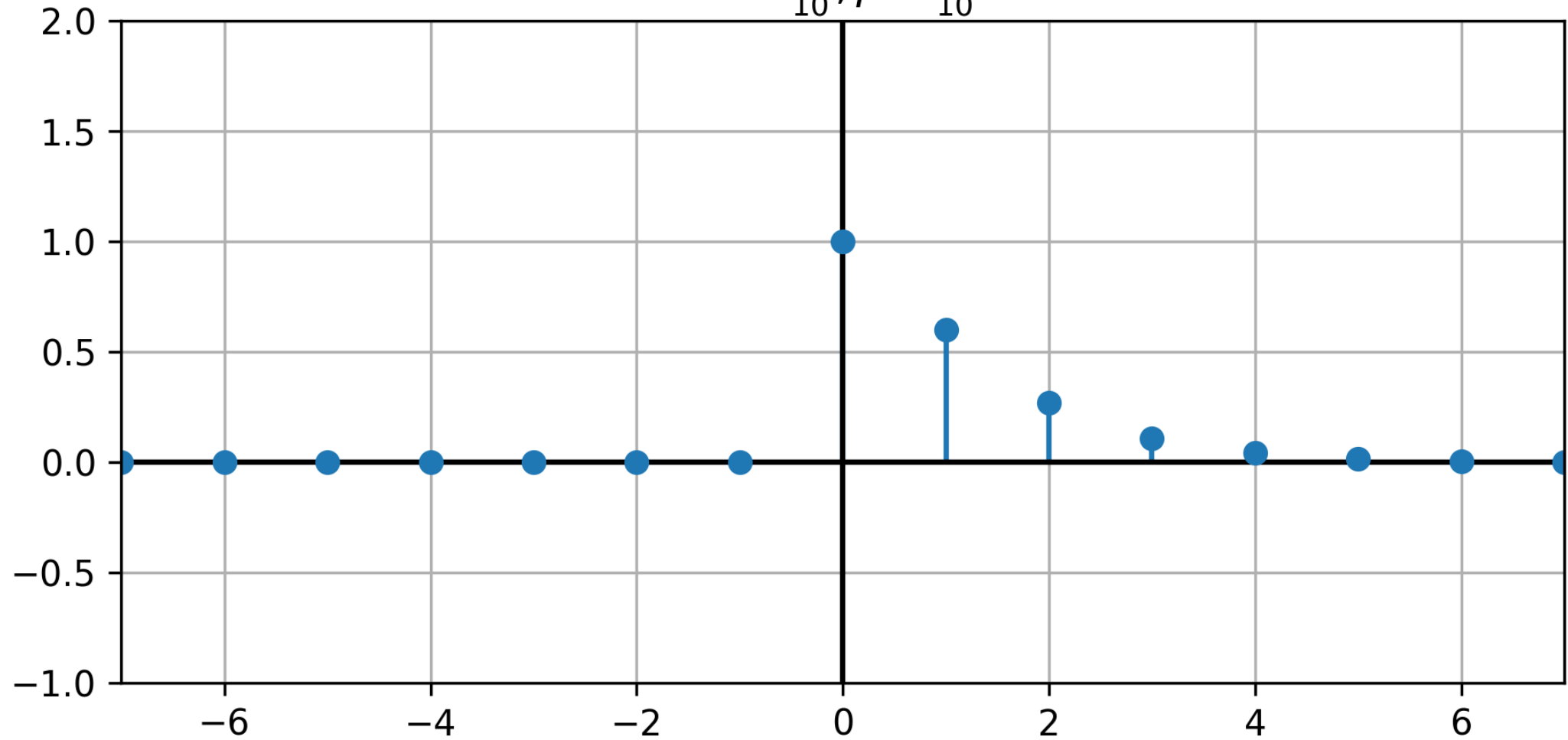
$$y[n] = \sum_{k=0}^n \alpha^k \alpha^n \alpha^{-k}$$

$$y[n] = \alpha^n \sum_{k=0}^n 1$$

$$y[n] = \alpha^n (n + 1) u[n]$$

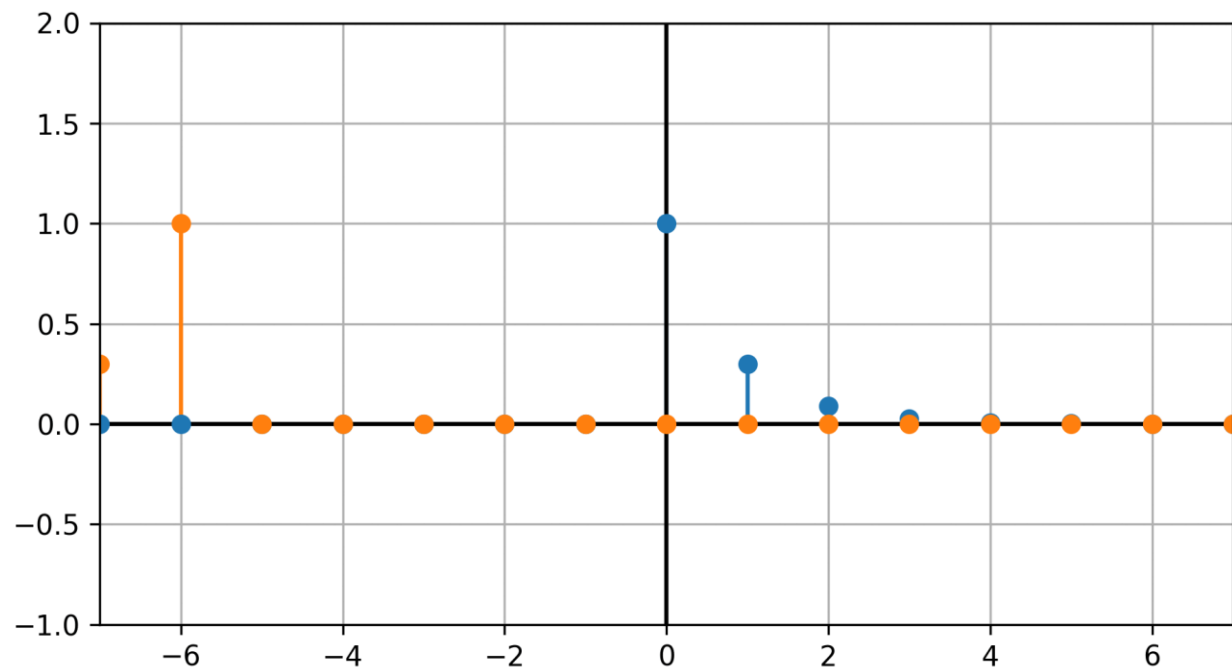
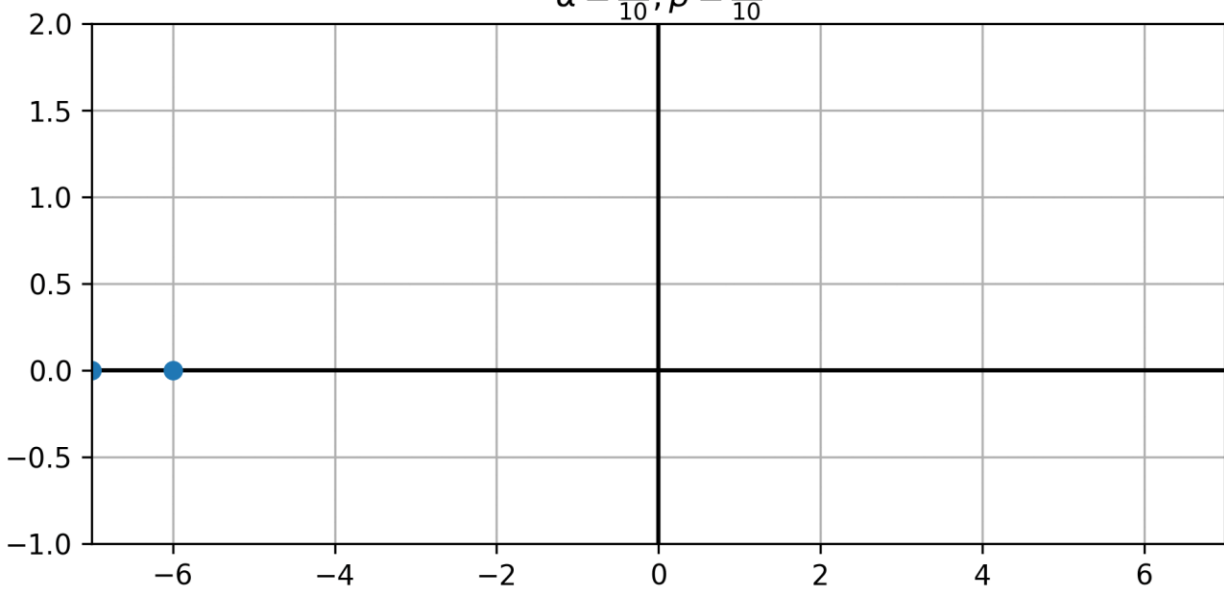
$$\alpha^n u[n] * \beta^n u[n]$$

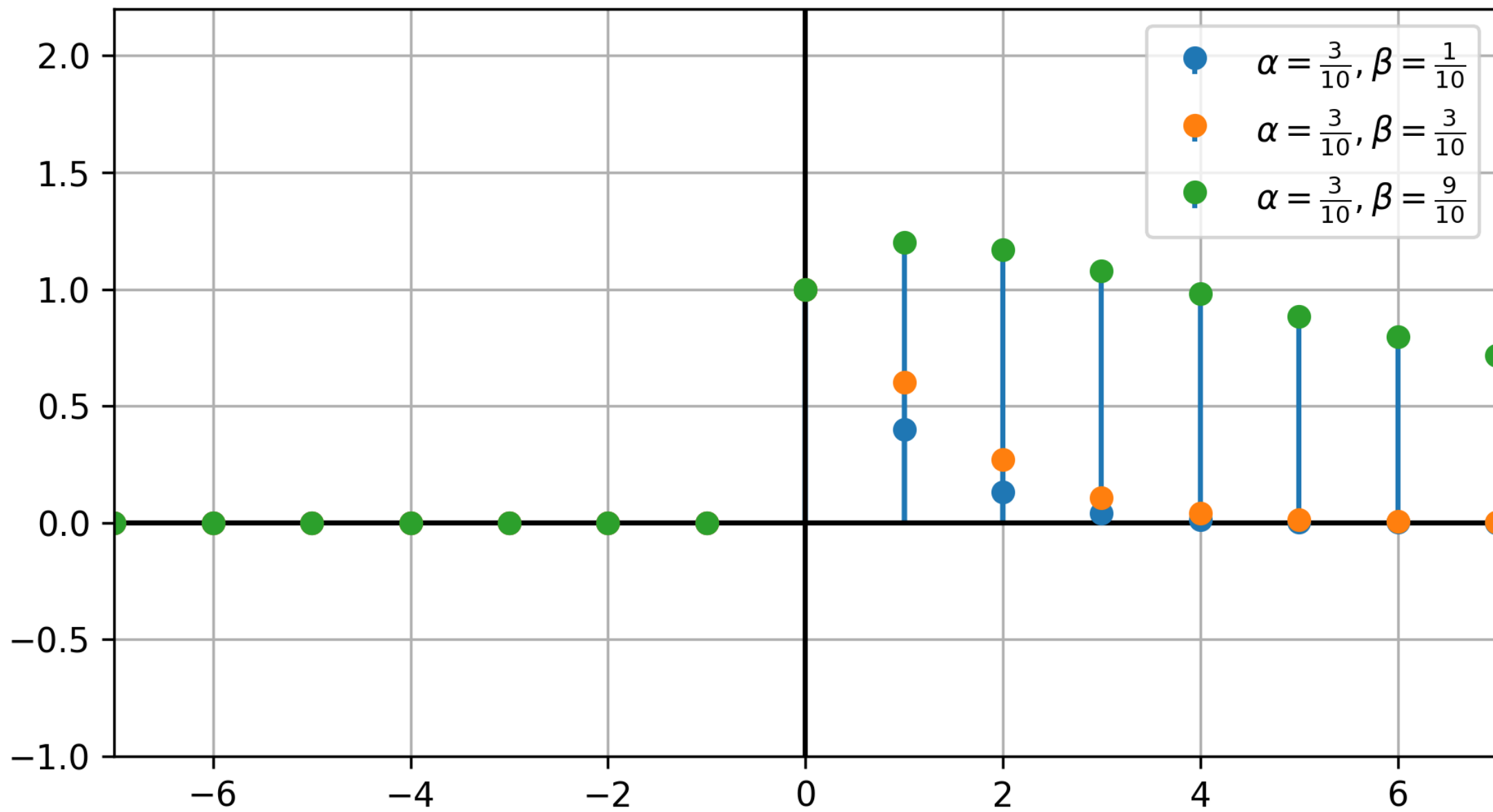
$$\alpha = \frac{3}{10}, \beta = \frac{3}{10}$$



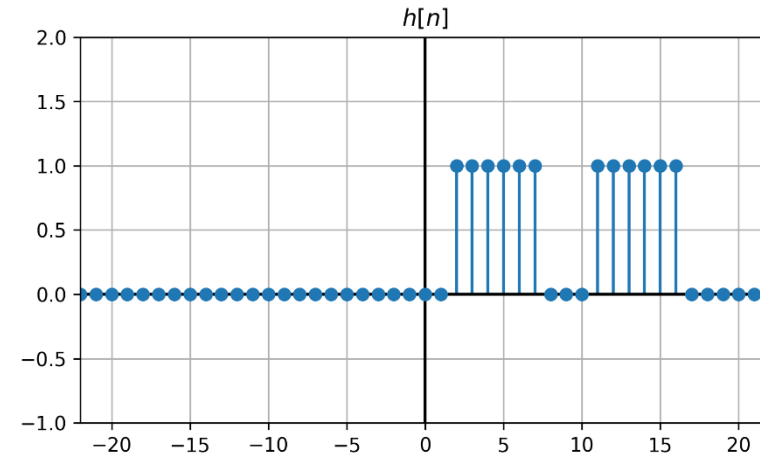
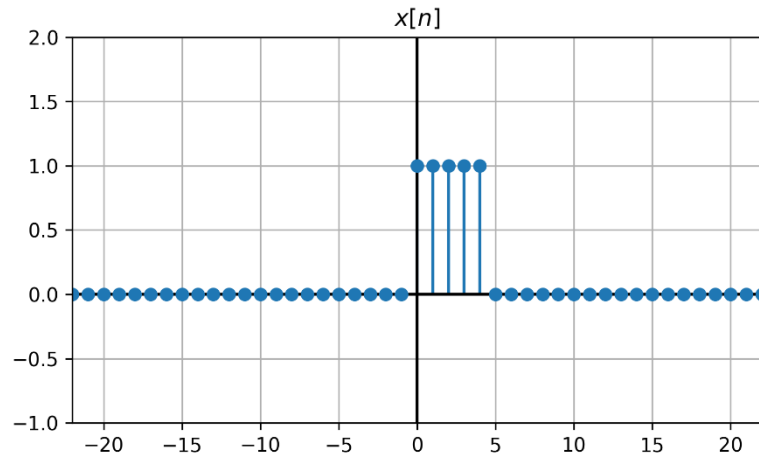
$$\alpha^n u[n] * \beta^n u[n]$$

$$\alpha = \frac{3}{10}, \beta = \frac{3}{10}$$





d) $x[n]$ y $h[n]$ son como en la figura:



$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

$$y[n] = x[0]h[n] + x[1]h[n - 1] + x[2]h[n - 2] + x[3]h[n - 3] + x[4]h[n - 4]$$

$$y[n] = h[n] + h[n - 1] + h[n - 2] + h[n - 3] + h[n - 4]$$

